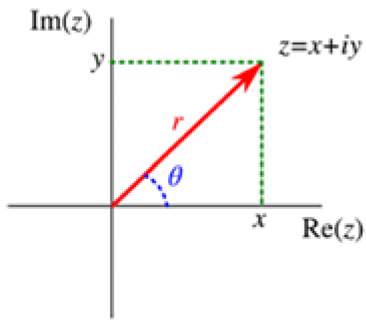
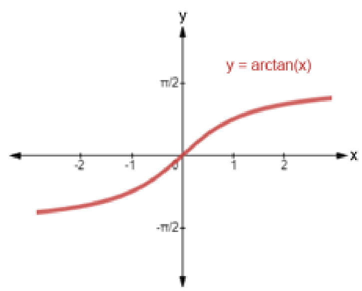
CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

1. Recall the Cartesian representation of a complex number z = reiθ and the polar representation z = reiθ. You already know that to convert polar to Cartesian representation, x = rcos θ and y = rsin θ.



   a. When converting z = x + yi to z = reiθ, obviously we have r = √x2 + y2 . However, the simplistic formula "θ = arctan(y/x)" is applicable only if x > 0, since "arctan" is valued in (–π/2, +π/2).



Therefore, please fill out the correct formula for $-\pi \le \theta < \pi$:

      i.

$$\theta = \begin{cases} \arctan(y/x), & x > 0 \\ ? - \arctan(y/|x|), & x < 0, y > 0 \\ ? - \arctan(y/|x|), & x < 0, y \le 0 \\ ?, & x = 0, y > 0 \\ ?, & x = 0, y < 0 \end{cases}$$

      ii. Analyze Each Case:
         1. When x > 0 (Quadrants I and IV):
            a. θ = arctan(y/x)
         2. When x < 0 and y > 0 (Quadrant II):
            a. θ = π + arctan(y/|x|)
         3. When x < 0 and y ≤ 0 (Quadrant III):
            a. θ = π + arctan(y/|x|)
         4. When x = 0 and y > 0 (positive y-axis):
            a. θ = π/2
         5. When x = 0 and y < 0 (negative y-axis):
            a. θ = -π/2
      iii. Complete Formula θ:
         1. arctan(y/x), for x > 0
         2. π + arctan(y/|x|), for x < 0, y > 0
         3. π + arctan(y/|x|), for x < 0, y ≤ 0

4. π/2, for x = 0, y > 0
5. -π/2, for x = 0, y < 0

b. Recall from the previous homework: |z1z2| = |z1|·|z2|, ∡(z1z2) = ∡z1 + ∡z2, and whenever z2 is nonzero, |z1/z2| = |z1|/|z2|, ∡(z1/z2) = ∡z1 – ∡z2. In particular, |z2| = |z|2, ∡(z2) = 2∡z, |1/z| = 1/|z|, ∡(1/z) = –∡z. Please find the polar representations of the following numbers (where a, b, c are positive real constants):

$$z_1 = \frac{1}{1+ai},\ \ z_2 = \frac{1-ai}{(1+bi)(1+ci)^2},\ \ z_3 = \frac{(1-bi)^3}{1-a^2+2aci}\text{(assuming } |a|<1),\ \ z_4 = \frac{(1-ai)(1+bi)}{(1+ai)^2}e^{b-ci}.$$

For example, $z_1 = r_1 \exp(i\theta_1)$ where $r_1 = 1/\sqrt{1+a^2}$ and $\theta_1 = -\arctan a$.

i.
ii. For z1 = 1/(1 + ai):
    1. r1 = 1/sqrt(1 + a^2)
    2. θ1 = -arctan(a)
    3. Therefore, z1 = [1/sqrt(1 + a^2)] * e^(-i*arctan(a))
iii. For z2 = (1 - ai)/[(1 + bi)(1 + ci)^2]:
    1. r2 = sqrt(1 + a^2)/[sqrt(1 + b^2)(1 + c^2)]
    2. θ2 = -arctan(a) - arctan(b) - 2arctan(c)
    3. Therefore, z2 = [sqrt(1 + a^2)/(sqrt(1 + b^2)(1 + c^2))] * e^(-i*(arctan(a) + arctan(b) + 2arctan(c)))
iv. For z3 = (1 - bi)^3/(1 - a^2 + 2aci), where |a| < 1:
    1. r3 = (1 + b^2)^(3/2)/sqrt((1 - a^2)^2 + 4a^2c^2)
    2. θ3 = -3arctan(b) - arctan(2ac/(1 - a^2))
    3. Therefore, z3 = [(1 + b^2)^(3/2)/sqrt((1 - a^2)^2 + 4a^2c^2)] * e^(-i*(3arctan(b) + arctan(2ac/(1 - a^2))))
v. For z4 = [(1 - ai)(1 + bi)/(1 + ai)^2] * e^(-ci):
    1. r4 = [sqrt(1 + a^2) * sqrt(1 + b^2)]/(1 + a^2)
    2. θ4 = -3arctan(a) + arctan(b) - c
    3. Therefore, z4 = [sqrt(1 + a^2) * sqrt(1 + b^2)/(1 + a^2)] * e^(-i*(3arctan(a) - arctan(b) + c))

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

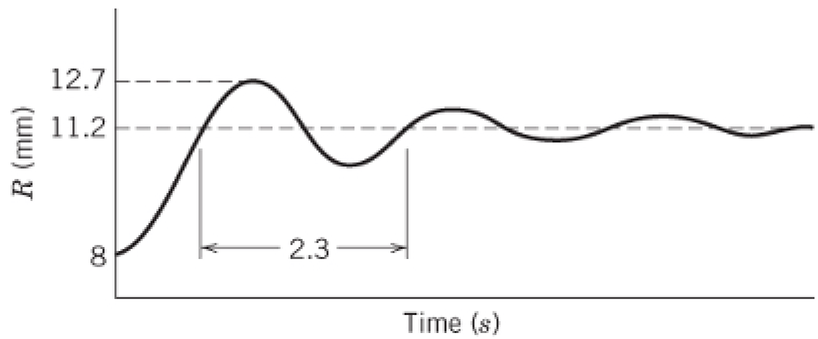
2. A heater for a semiconductor wafer has first-order dynamics, that is, the transfer function relating changes in temperature T to changes in the heater input power level P is where K has units [°C/kW] and τ has units [min]. The process is at steady state when an engineer changes the power input stepwise from 1 to 2 kW. She notes the following: $\frac{T'(s)}{P'(s)} = \frac{K}{\tau s+1}$, The process temperature initially is 100 °C. • 4 minutes after changing the power input, the temperature is 400 °C. • 30 minutes later the temperature is 500 °C.
    a. What are K and τ in the process transfer function?
        i. Calculate Process Gain (K):
            1. Initial temperature = 100°C
            2. Final temperature = 500°C
            3. Temperature change (ΔT) = 500°C - 100°C = 400°C
            4. Power change (ΔP) = 2 kW - 1 kW = 1 kW
            5. $K = \Delta T/\Delta P$ = 400°C/1 kW = 400°C/kW
        ii. Calculate Time Constant (τ):
            1. Using $T(t) = T_{final} - (T_{final} - T_{initial})e^{(-t/\tau)}$
            2. At t = 4 minutes, T = 400°C
            3. $400 = 500 - (500 - 100)e^{(-4/\tau)}$
            4. $400 = 500 - 400e^{(-4/\tau)}$
            5. $400e^{(-4/\tau)} = 100$
            6. $e^{(-4/\tau)} = 0.25$
            7. $-4/\tau = \ln(0.25)$
            8. $\tau = -4/\ln(0.25) = 2.89$ minutes
        iii. Final Values:
            1. K = 400°C/kW
            2. τ = 2.89 minutes
    b. If at another time the engineer changes the power input linearly at a rate of 0.5 kW/min, what can you say about the maximum rate of change of process temperature: When will it occur? How large will it be?
        i. Rate of Change Formula:
            1. $dT/dt = (K/\tau)(dP/dt)e^{(-t/\tau)}$
            2. Power rate of change = 0.5 kW/min
            3. Substituting values:
                a. $dT/dt = (400/2.89)(0.5)e^{(-t/2.89)}$
                b. $dT/dt = 69.24e^{(-t/2.89)}$
        ii. Rate of temperature changed
            1. $dT/dt = (K/\tau)(dP/dt)e^{(-t/\tau)}$
            2. dP/dt = 0.5 kW/min
            3. Maximum rate occurs at t = 0 when $e^{(-t/\tau)} = 1$
            4. Maximum rate = (400/2.89)(0.5) = 69.2°C/min
        iii. Final Values:
            1. Occurs immediately (t = 0)
            2. Value = 69.2°C/min

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

3. Second-order system considered as parallel connection of two first-order systems: Transfer function: $G(s) = Y(s)/U(s) = 10/[(5s+1)(3s+1)]$ Find $y(t \rightarrow \infty)$ for:
    a. Step input of height M:
        i. Input in Frequency Domain:
        ii.
            1. A step input of height M can be represented in the Laplace domain as:
                a. $U(s) = M/s$
            2. Output in the Laplace Domain:
                a. Using the transfer function:
                    i. $Y(s) = G(s) \cdot U(s) = [10/(5s+1)(3s+1)] \cdot [M/s]$
                    ii. So:
                    iii. $Y(s) = 10M/[s(5s+1)(3s+1)]$
                b. Partial Fraction Decomposition:
                    i. Decompose Y(s):
                        1. $10/[(5s+1)(3s+1)] = A/(5s+1) + B/(3s+1)$
                    ii. Multiply through by (5s+1)(3s+1):
                        1. $10 = A(3s+1) + B(5s+1)$
                    iii. Expand:
                        1. $10 = 3As + A + 5Bs + B$
                    iv. Group terms:
                        1. $10 = (3A + 5B)s + (A + B)$
                    v. Coefficients:
                        1. For s: $3A + 5B = 0$
                        2. For constant: $A + B = 10$
                    vi. Substitute into $3A + 5B = 0$:
                        1. $3A + 5(10 - A) = 0$
                        2. $3A + 50 - 5A = 0$
                        3. $-2A + 50 = 0$
                        4. $A = 25$
                    vii. Substitute $A = 25$ into $A + B = 10$:
                        1. $25 + B = 10$
                        2. $B = -15$
                    viii. Partial fraction decomposition is:
                        1. $10M/[s(5s+1)(3s+1)] = A/s + B/(5s+1) + C/(3s+1)$
                c. Finding the Steady-State Output:
                    i. final value theorem:
                        1. $y(t \rightarrow \infty) = \lim[s \rightarrow 0]\, sY(s)$
                    ii. Plugging into our equation:
                        1. $y(t \rightarrow \infty) = \lim[s \rightarrow 0]\, s \cdot [10M/s(5s+1)(3s+1)] = 10M/[(5 \cdot 0+1)(3 \cdot 0+1)] = 10M/(1 \cdot 1) = 10M$
                d. $y(t \rightarrow \infty) = 10M$

b. Unit impulse input δ(t):
   1. The unit impulse input in the s-domain is U(s) = 1
   2. The output in the s-domain is:
      a. Y(s) = G(s)U(s) = 10/[(5s+1)(3s+1)] × 1
      b. Y(s) = 10/[(5s+1)(3s+1)]
   3. Using inverse Laplace transform of 1/(as+1) = e^(-t/a):
      a. y(t) = 25e^(-t/5) - 15e^(-t/3)
   4. y(t→∞) = 0

4. Underdamped Second-Order System [Seborg 5.14]



a. Calculate all important parameters and write an approximate transfer function in the form: R'(s)/P'(s) = K/(τ^2s^2 + 2ζτs + 1) where: R' is the instrument output deviation (mm), P' is the actual pressure deviation (psi).
   i. Calculate Gain (K):
      1. ΔP = 31 psi - 15 psi = 16 psi
      2. ΔR = 11.2 mm - 8 mm = 3.2 mm
      3. K = ΔR / ΔP = 3.2 mm / 16 psi = 0.2 mm/psi
   ii. Calculate Overshoot (OS):
      1. OS = (Peak Value - Final Value) / Final Value
      2. OS = (12.7 mm - 11.2 mm) / (11.2 mm - 8) = 1.5/3.2 = 0.46875
   iii. Calculate Damping Coefficient (ζ):
      1. OS = exp(-πζ / sqrt(1 - ζ^2))
      2. 0.46875 = exp(-πζ / sqrt(1 - ζ^2))
      3. ln(0.46875) = -πζ / sqrt(1 - ζ^2)
      4. -0.7576 = -πζ / sqrt(1 - ζ^2)
      5. 0.7576/π = ζ / sqrt(1-ζ^2). Let's name 0.7576/π as C
      6. C = ζ / sqrt(1-ζ^2)
      7. C^2 = ζ^2/(1-ζ^2)
      8. C^2 - C^2 * ζ^2 = ζ^2
      9. C^2 = ζ^2 * (1+C^2)
      10. ζ = C/sqrt(1+C^2)
      11. ζ = 0.234
   iv. Calculate Time to first peak (Tp):
      1. Given as 2.3
   v. Calculate damped period (Td):

1. $T_d = 2 * T_p = 4.6$ s

vi. Calculate Natural Period (τ):

1. $T_d = 2\pi\tau/\sqrt{1-\zeta^2}$
2. $\tau = T_d\sqrt{1-\zeta^2}/(2\pi) = 4.6\sqrt{1-0.234^2}/(2\pi) \approx 0.712$ s
3. $\tau^2 = 0.507$

vii. Assemble Transfer Function:

1. $R'(s)/P'(s) = K / (\tau^2 s^2 + 2\zeta\tau s + 1) = 0.2/(0.507s^2 + 2 * 0.234 * 0.712s + 1)$
2. $= 0.2/(0.507s^2 + 0.333s + 1)$

b. Write an equivalent differential equation model in terms of actual (not deviation) variables.

i. From Transfer Function:

1. $0.507s^2R'(s) + 0.333sR'(s) + R'(s) = 0.2P'(s)$

ii. Inverse Laplace Transform:

1. $0.507(d^2R'/dt^2) + 0.333(dR'/dt) + R' = 0.2P'$

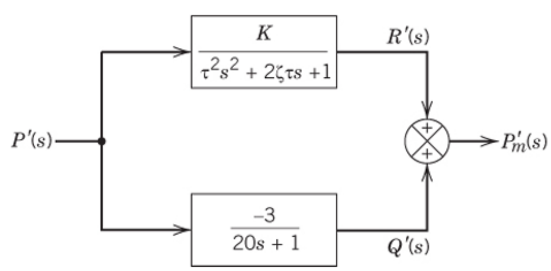
iii. Use actual variables:

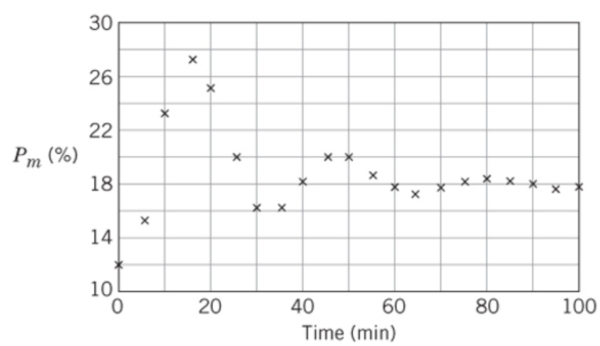
1. $R' = R - 8$
2. $P = P' + 15$
3. $P' = P – 15$

iv. Substitute and Simplify:

1. $0.507(d^2R/dt^2) + 0.333(dR/dt) + (R-8) = 0.2(P-15)$
2. $0.507(d^2R/dt^2) + 0.333(dR/dt) + R - 8 = 0.2P - 3$
3. $0.507(d^2R/dt^2) + 0.333(dR/dt) + R = 0.2P + 5$

5. High-order dynamics due to connection of multiple simpler low-order models [Seborg 6.7], A pressure-measuring device analyzed and modeled Device responds to pressure changes as a first-order process in parallel with a second-order process First-order process characteristics: Gain = -3, Time constant = 20 min, Additional test data: Step change in P from 4 to 6 psi, Actual output Pm (not Pm') plotted in figure.





a. Determine Q'(s)

i. From the block diagram:

1. The first-order process model for Q'(s) is given as: $Q'(s) = (-3)/(20s + 1)P'(s)$.
2. $Q'(s) = [-3P'(s)]/(20s + 1)$

b. Estimate the values of: K (gain), τ (time constant), ζ (damping coefficient)

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

i. K (gain):
   1. The second-order process transfer function is:
      a. $R'(s) = K/(\tau^2 s^2 + 2\zeta\tau s + 1)P'(s)$
   2. The steady-state output Pm happens when s = 0.
      a. $G(s) = -3/(20s + 1) + K/(\tau^2 s^2 + 2\zeta\tau s + 1$
   3. For steady state (s = 0):
      a. $Q'(s) = -3/1$ and $R'(s) = K$
   4. Output Pm is the summation of Q'(s) and R'(s):
      a. $Pm = Q'(s) + R'(s)$
   5. From the graph:
      a. Input step change: $\Delta P = 2$ psi (from 4 to 6 psi)
      b. Output response (ΔPm) reaches ~22%
   6. Solve for K:
      a. $\Delta Pm = Q'(s) + R'(s) = -3 \times 2 + K \times 2 = 22$
      b. $-6 + 2K = 22 \rightarrow K = 14$

ii. ζ (damping coefficient):
   1. The damped natural frequency (ωd) is related to tp as:
      a. $tp = \pi/\omega d$
      b. $\omega d = \pi/tp = \pi/15 \approx 0.2094$ rad/min
   2. The natural frequency (ωn) is related to ωd as:
      a. $\omega n = \omega d/\mathrm{sqrt}(1 - \zeta^2)$
   3. Damping ratio ζ
      a. Using the overshoot (Mp) from the plot:
         i. Mp = (Peak value - Steady-state value)/(Steady-state value) = (26 - 22)/22 = 0.1818
      b. The overshoot (Mp) is related to ζ as:
         i. $Mp = \exp(-\pi\zeta/\mathrm{sqrt}(1-\zeta^2))$
      c. Taking the natural logarithm:
         i. $\ln(0.1818) = -\pi\zeta/\mathrm{sqrt}(1-\zeta^2)$
         ii. $-1.705 = -\pi\zeta/\mathrm{sqrt}(1-\zeta^2)$
      d. Solving for ζ:
         i. $\zeta = 0.412$
   4. τ (time constant):
      a. $\tau = 1/\omega n$
      b. $\omega n = \omega d/\mathrm{sqrt}(1-\zeta^2) = 0.2094/\mathrm{sqrt}(1-(0.412)^2) \approx 0.2284$
      c. Solve for τ:
         i. $\tau = 1/\omega n \approx 1/0.2284 \approx 4.38$ min

c. What is the overall transfer function for the measurement device Pm'(s)/P'(s)?
   i. $Pm'(s)/P'(s) = -3/(20s + 1) + 14/(\tau^2 s^2 + 2\zeta\tau s + 1)$
      1. Substitute $\tau = 4.38$ and $\zeta = 0.412$:
         a. $\tau^2 = (4.38)^2 = 19.18$
         b. $2\zeta\tau = 2 \times 0.412 \times 4.38 \approx 3.61$
         c. $Pm'(s)/P'(s) = -3/(20s + 1) + 14/(19.18s^2 + 3.61s + 1)$

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

6. Second-order dynamics due to recycle connections. [Seborg 6.19]. Consider the following connection of isothermal reactors. A first-order reaction A → B occurs in both reactors, with reaction rate constant k. The volumes of liquid in the reactors, V, are constant and equal; the flow rates F_0, F_1, F_2 and R are constant. Assume constant physical properties and a negligible time delay for the recycle line.
    a. Write a mathematical model for this process.
        i. First Reactor:
            1. Rate of accumulation = Inflow - Outflow - Reaction rate
            2. $V(dC_1/dt) = F_0C_0 - F_1C_1 - kVC_1$
        ii. Simplify:
            1. $dC_1/dt = (F_0C_0)/V - (F_1C_1)/V - kC_1$
        iii. Second Reactor:
            1. Rate of accumulation = Inflow - Outflow - Reaction rate
            2. $V(dC_2/dt) = F_1C_1 - F_2C_2 - kVC_2$
        iv. Simplify:
            1. $dC_2/dt = (F_1C_1)/V - (F_2C_2)/V - kC_2$
        v. Equations:
            1. $dC_1/dt = (F_0C_0)/V - (F_1C_1)/V - kC_1$
            2. $dC_2/dt = (F_1C_1)/V - (F_2C_2)/V - kC_2$
    b. Derive a transfer function model relating the output concentration of A, C_2 to the inlet concentration of A, C_0.
        i. Laplace Transform of the Material Balance Equations
        ii. For reactor 1:
            1. $VsC_1(s) = F_0C_0(s) - F_1C_1(s) - kVC_1(s)$
        iii. Solve for $C_1(s)$:
            1. $(Vs + F_1 + kV)C_1(s) = F_0C_0(s)$
            2. $C_1(s) = F_0/(Vs + F_1 + kV)C_0(s)$
        iv. For reactor 2:
            1. $VsC_2(s) = F_1C_1(s) - F_2C_2(s) - kVC_2(s)$
        v. Solve for $C_2(s)$:
            1. $(Vs + F_2 + kV)C_2(s) = F_1C_1(s)$
        vi. Substitute $C_1(s) = F_0/(Vs + F_1 + kV)C_0(s)$:
            1. $C_2(s) = [F_1/(Vs + F_2 + kV)][F_0/(Vs + F_1 + kV)]C_0(s)$
        vii. Simplify:
            1. $C_2(s) = F_0F_1/[(Vs + F_1 + kV)(Vs + F_2 + kV)]C_0(s)$
        viii. Transfer function:
            1. $C_2(s)/C_0(s) = F_0F_1/[(Vs + F_1 + kV)(Vs + F_2 + kV)]$
    c. Verify that, in the limit of no recycle (R → 0), the transfer function derived in (b) is equivalent to the transfer function of the two tanks connected in series.
        i. When R → 0:
            1. $F_1 = F_2$, meaning there is no recycle flow.
        ii. The simplified transfer function becomes:
            1. $C_2(s)/C_0(s) = F_0F_1/[(Vs + F_1 + kV)(Vs + F_1 + kV)]$

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

d. Show that when k = 0 and a very large recycle flow rate is used (i.e., the limit as R → ∞), the transfer function derived in (b) becomes the transfer function of a single tank that has the volume equal to 2V and a gain of 1. Hint: Recognize that $F_1 = R + F_2$ and $F_0 = F_2$.
    i. When k = 0:
        1. No chemical reaction occurs, so C_2 depends entirely on flow and mixing.
    ii. When R → ∞:
        1. For very large recycle flow, $F_1 >> F_2$, and the system behaves as if the two reactors are a single tank with a total volume of 2V.
    iii. The transfer function for a single tank of volume 2V is:
        1. $C_2(s) = C_0(s)/(2Vs)$
    iv. Gain of such a system = 1

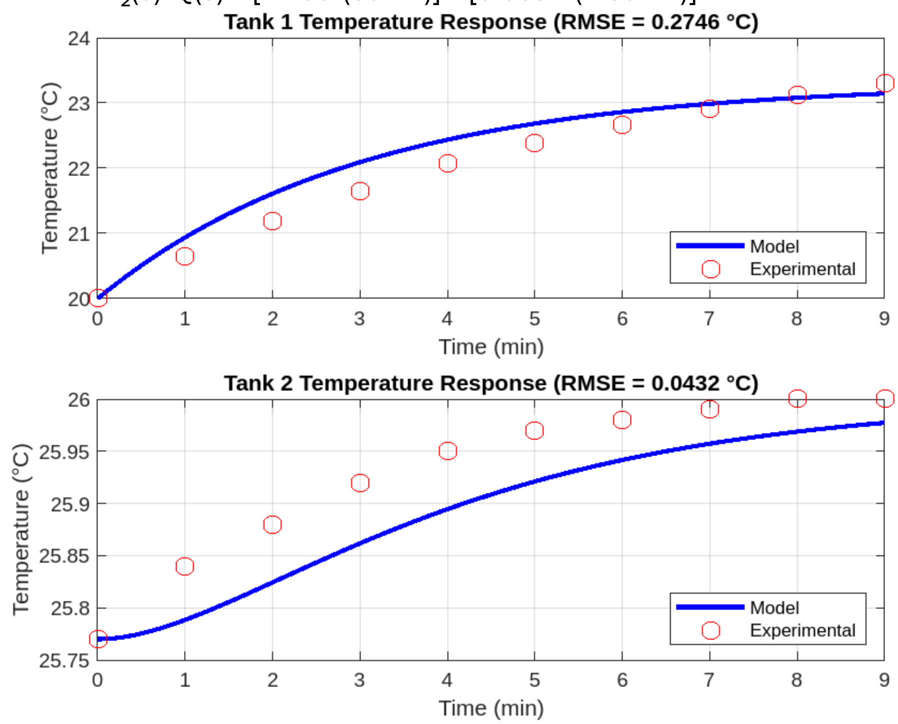
CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

7. In industrial practice, engineers often do not build models from first principles, but instead put a step signal (or other designed signals) on the input and "identify" the model from the resulting output response data. This procedure, called "system identification", can be done either by engineer's flesh-eye readings, rough calculations, or statistical software. Matlab and Python have well established toolboxes for it. On the other hand, since system identification is an extremely time-consuming (and hence expensive) task, it remains a focus of today's industrial control technology research and development. To answer the following question, you may want to read Sections 7.1 and 7.2 of Seborg first. A process consists of two stirred tanks with input q and outputs T_1 and T_2 (see figure below). To test the hypothesis that the dynamics in each tank are basically first-order, a step change in q is made from 82 to 85 L/min, with output responses given in the table.
    a. Find the transfer functions $T_1(s)/Q(s)$ and $T_2(s)/T_1(s)$. Assume that they are of the form $K/(\tau s + 1)$.
        i. Input: q (flow rate)
        ii. Outputs: $T_1(t)$ (temperature response from Tank 1) and $T_2(t)$ (temperature response from Tank 2)
        iii. Dynamics in both tanks are assumed first-order:
            1. Tank 1: $T_1(s)/Q(s) = K_1/(\tau_1 s + 1)$
            2. Tank 2: $T_2(s)/T_1(s) = K_2/(\tau_2 s + 1)$
        iv. Using the general first-order system response to a step input:
            1. T(t) = Tfinal + (Tinitial - Tfinal)e^(-t/τ)
        v. Analyze the Tank 1 response
            1. From the table, the response of $T_1(t)$
            2. Steady-state gain (K_1):
                a. The step change in q is $\Delta q = 85 - 82 = 3$ L/min
                b. The steady-state change in T_1 is $\Delta T_1 = 23.31 - 20.00 = 3.31$ °C
                c. $K_1 = \Delta T_1/\Delta q = 3.31/3 = 1.103$ °C/L/min
            3. Time constant (τ_1):
                a. From the data, $T_1(t)$ reaches 63.2% of its total change by t = 3 min
                b. $\tau_1 = 3$ min
                c. The transfer function for Tank 1 is:
                d. $T_1(s)/Q(s) = 1.103/(3s + 1)$
        vi. Analyze the Tank 2 response
            1. From the table, the response of $T_2(t)$
            2. Steady-state gain (K_2):
                a. The steady-state change in T_2 is $\Delta T_2 = 26.00 - 25.77 = 0.23$ °C
                b. $K_2 = \Delta T_2/\Delta T_1 = 0.23/3.31 = 0.0694$
            3. Time constant (τ_2):
                a. From the data, $T_2(t)$ reaches 63.2% of its total change by t = 1.5 min
                b. $\tau_2 = 1.5$ min
            4. The transfer function for Tank 2 is:

CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

a. $T_2(s)/T_1(s) = 0.0694/(1.5s + 1)$

b. Calculate the model responses to the same step change in q and plot with the experimental data.
   i. Tank 1 Model:
      1. Input $Q(s)$, transfer function $T_1(s)/Q(s) = 1.103/(3s + 1)$
   ii. Tank 2 Model:
      1. Input $T_1(s)$, transfer function $T_2(s)/T_1(s) = 0.0694/(1.5s + 1)$
   iii. The overall response is:
      1. $T_2(s)/Q(s) = [1.103/(3s + 1)] \times [0.0694/(1.5s + 1)]$
   iv.



CHE 435 - Process Systems Analysis and Control
Spring 2025 – Dr. Tang
Brennon Brown
HW3

8. Process control in depth: Lead-lag dynamics [Seborg 6.4] Consider a process model (for simplicity, you may assume K = 1): $Y(s)/U(s) = K(\tau_2 s + 1)/(\tau_1 s + 1)(\tau_2 s + 1)$, ($\tau_1 > \tau_2$) For a step input, prove that:
    a. y(t) can exhibit an extremum (maximum or minimum value) in the step response only if: $(1 - \tau_2/\tau_1)/(1 - \tau_2/\tau_1) > 1$
        i. Write the step response:
        ii. Take Y(s) and separate it using partial fraction expansion:
            1. $Y(s) = 1/s - (\tau_1 - \tau_2)/\tau_1 \times 1/(\tau_1 s + 1)$
        iii. Transform back to the time domain:
            1. $y(t) = 1 - [(\tau_1 - \tau_2)/\tau_1]e^{\wedge}(-t/\tau_1) - (\tau_2/\tau_1)e^{\wedge}(-t/\tau_2)$
        iv. Differentiate y(t):
            1. $dy(t)/dt = [(\tau_1 - \tau_2)/\tau_1] \times (1/\tau_1)e^{\wedge}(-t/\tau_1) + (\tau_2/\tau_1) \times (1/\tau_2)e^{\wedge}(-t/\tau_2)$
        v. Simplify:
            1. $dy(t)/dt = [(\tau_1 - \tau_2)/(\tau_1\tau_1)]e^{\wedge}(-t/\tau_1) - (\tau_2/\tau_1\tau_2)e^{\wedge}(-t/\tau_2)$
        vi. Set dy(t)/dt = 0 to find the condition for extrema:
            1. $[(\tau_1 - \tau_2)/\tau_1^2]e^{\wedge}(-t/\tau_1) = (1/\tau_1)e^{\wedge}(-t/\tau_2)$
        vii. Cancel terms and rearrange:
            1. $(\tau_1 - \tau_2)/\tau_1 = e^{\wedge}[t(1/\tau_2 - 1/\tau_1)]$
        viii. Taking the natural logarithm:
            1. $t = [\tau_1\tau_2/(\tau_1 - \tau_2)]\ln[(\tau_1 - \tau_2)/\tau_1]$
        ix. For the extremum
            1. $(1 - \tau_2/\tau_1)/(1 - \tau_2/\tau_1) > 1$
    b. Overshoot occurs only for $\tau_2/\tau_1 > 1$
        i. Overshoot occurs
            1. $y(t) = 1 - [(\tau_1 - \tau_2)/\tau_1]e^{\wedge}(-t/\tau_1) - (\tau_2/\tau_1)e^{\wedge}(-t/\tau_2)$
        ii. For y(t) to exceed 1
            1. $\tau_2/\tau_1 > 1$, i.e., $\tau_2 > \tau_1$
                a. we are given $\tau_1 > \tau_2$, so overshoot does not occur unless this condition is violated.
    c. Inverse response occurs (namely an extreme of y(t) has a different sign from K) only for $\tau_2 < 0$.
        i. From the step response:
            1. $y(t) = 1 - [(\tau_1 - \tau_2)/\tau_1]e^{\wedge}(-t/\tau_1) - (\tau_2/\tau_1)e^{\wedge}(-t/\tau_2)$
        ii. Initially, at t = 0:
            1. $y(0) = 1 - (\tau_1 - \tau_2)/\tau_1 - \tau_2/\tau_1 = 1 - 1 = 0$
        iii. For an inverse response to occur, set $\tau_2 < 0$. The inverse response exists only when $\tau_2 < 0$.
    d. If an extremum in y exists, the time at which it occurs can be found analytically. What is it?
        i. The time at which the extremum occurs is:
            1. $t = [\tau_1\tau_2/(\tau_1 - \tau_2)]\ln[(\tau_1 - \tau_2)/\tau_1]$